

Module 02: The Crocheter as Neural Architect

Learning Objectives

1. Understand the crocheter as an agent who designs, selects, and physically builds network architectures in yarn — and recognize that the crocheter's own brain is a neural network designing another network.
2. Compare the crocheter's stitch selection and tension control to neural network weight assignment, activation functions, and parameter tuning.
3. Recognize how each stitch placement is a decision node governed by Active Inference — the crocheter minimizes surprise by selecting stitches that match the generative model (the pattern and the desired fabric).

Introduction

In the previous module, we looked at crocheted fabric as a topological surface and a network graph. We examined the structure of the thing being made. Now we turn to the maker.

You, the crocheter, are a neural architect. Not metaphorically — literally. Every time you sit down with hook and yarn, you are designing and constructing a physical network. You decide the architecture (the pattern), select the connection types (the stitches), adjust the connection strengths (your tension), and execute the build node by node, stitch by stitch. The finished fabric is a network you have built with your hands, and every choice you made along the way shaped its topology.

Here is the twist that makes this fascinating: the tool you use to design this network — your brain — is itself a neural network. A neural network is building a neural network. The architect is made of the same stuff as the architecture. This recursive loop, where an agent made of networks builds networks, is at the heart of Active Inference's view of agency. The crocheter does not simply follow instructions. The crocheter is a prediction machine, constantly generating expectations about what the next stitch should feel like, look like, and do — and then acting to make those predictions come true.

Key Concepts

1. The Crocheter as Network Designer

Every pattern is a blueprint for a network. When a crocheter reads "Row 1: Ch 20. Sc in 2nd ch from hook and in each ch across. (19 sc)," they are reading an architectural specification. The pattern tells you how many nodes to create (19 stitches), what type of connections to use (single crochet), and how they link to the previous layer (into each chain).

But the crocheter is more than a builder following a blueprint. Experienced crocheters make **architectural decisions** constantly — decisions that go beyond what the pattern specifies. They choose:

The base architecture. Will this be worked in rows (feedforward) or in the round (recurrent)? Will it have a fixed width (rectangular, like a scarf) or a changing width (shaped, like a garment piece)? These are global architectural choices that determine the network's overall topology.

The stitch type. Different stitches create fundamentally different connection patterns. A **single crochet** (sc) is compact: the hook enters the stitch below, yarn over, pull through, yarn over, pull through both loops. The resulting connection is tight, dense, and short. A **double crochet** (dc) is taller: yarn over, insert hook, pull through, yarn over, pull through two, yarn over, pull through two. It creates a longer, more open connection. A **treble crochet** (tr) is taller still.

In neural network terms, these stitch types function like different **activation functions**. An activation function determines how a neuron transforms its input into output. A tight single crochet is like a step function — it transmits structure directly and compactly. A tall double crochet is like a ReLU (rectified linear unit) — it allows more signal to pass through, with more vertical reach. A chain space (skipping stitches with chains) is like a dropout layer — it intentionally leaves gaps in the connectivity, creating openness and reducing density.

The crocheter choosing between sc, dc, and tr is making the same kind of decision as a machine learning engineer choosing between activation functions. Both choices shape how information flows through the network.

The connectivity pattern. Beyond individual stitch types, the crocheter decides how stitches connect to the previous layer. Working into both loops (standard), working into the front loop only (FLO), or working into the back loop only (BLO) creates different connection patterns. FLO and BLO leave one loop unworked, creating a visible ridge and a different structural bond — like a neural network where certain connections are deliberately pruned.

Shell stitches (multiple stitches in one stitch) create fan-out connections — one node in the previous layer connects to five or more in the current layer. Decrease stitches (working two or more stitches together) create fan-in connections — multiple nodes in the previous layer merge into one. These are the same operations that neural network architects use to change the dimensionality of layers: expanding layers have more fan-out, contracting layers have more fan-in.

The crocheter is, stitch by stitch, deciding the dimensionality of each layer and the connectivity between layers. This is network architecture design, executed in real time, with yarn.

2. Stitch Selection as Weight Assignment

In a neural network, each connection between neurons has a **weight** — a numerical value that determines how strongly one neuron influences another. High weights mean strong influence; low weights mean weak influence. The process of training a neural network is, fundamentally, the process of adjusting these weights until the network produces the desired output.

In crochet, the analog of weight is **tension** — combined with stitch type and yarn characteristics.

Tension as a continuous weight parameter. Every crocheter has a characteristic tension: how tightly they hold the yarn, how snugly they pull each loop. Tight tension creates dense, stiff fabric with strong connections between stitches — high weights. Loose tension creates drapery, flowing fabric with softer connections — lower weights. The same pattern, crocheted by two people with different tension, produces two different fabrics. The network architecture is the same; the weights are different.

This is exactly what happens in neural networks. Two networks with identical architecture (same number of layers, same connectivity) can produce entirely different outputs if their

weights are different. The weights are what give the network its particular character, its specific behavior. In crochet, tension is the weight parameter, and it is continuous — not just "tight" or "loose," but an infinitely variable spectrum that experienced crocheters control with remarkable precision.

Stitch height as connection strength. Taller stitches create longer connections. A single crochet connects tightly to the stitch directly below it. A double crochet reaches higher, creating a more extended link. A treble crochet reaches higher still. The physical height of the stitch determines the "reach" of the connection — how far the signal travels. In neural network terms, taller stitches have different receptive fields. A treble crochet "sees" more of the surrounding structure than a single crochet, just as a convolution with a larger kernel size captures more of the input.

Learning to adjust weights. When a beginning crocheter practices gauge swatches, they are training their own neural network (their brain and hands) to produce consistent weights (tension). The first swatch might be too tight, the second too loose. Gradually, the crocheter's motor system converges on the target tension — just as a neural network's weights converge during training through gradient descent. The crocheter's "gradient" is the difference between their current gauge and the target gauge. Each practice swatch updates their internal model.

This process is Active Inference in action. The crocheter has a **generative model** that predicts what a given tension should produce (a certain number of stitches per inch). When the swatch does not match the prediction, the crocheter experiences **prediction error** — free energy that needs to be minimized. The crocheter can minimize this error in two ways: adjust their tension (active inference — changing the world to match the model) or adjust their expectation (perceptual inference — changing the model to match the world, perhaps by switching to a different hook size).

3. Agency at the Node Level

Zoom in to a single moment in the crochet process. Your hook is loaded. You are about to make the next stitch. Where do you insert the hook? Into the next stitch? Into the one after that (skipping a stitch)? Into both loops or just the front loop? Do you yarn over once (sc), twice (dc), or three times (tr)? Do you make the stitch at all, or do you chain over it?

This single moment — hook poised over the fabric — is a **decision node**. The crocheter has a set of available actions (the possible stitches and insertion points), and must select one. In a simple pattern, the choice is prescribed: "sc in next st." But even in prescribed patterns, micro-decisions abound. Exactly where the hook tip enters the stitch. How much yarn is pulled through. Whether the loop is snugged up or left relaxed.

A crocheter working a row of 200 stitches makes at least 200 explicit decisions and hundreds more implicit micro-decisions. Over the course of a project, the number of decision nodes runs into the thousands or tens of thousands. Each one shapes the network.

Active Inference at each node. At every decision node, the crocheter is running a miniature Active Inference cycle:

1. **Prediction.** The generative model (the pattern, the crocheter's experience, the feel of the fabric) predicts what the next stitch should be. "The next stitch should be a single crochet in the next stitch, and it should feel like a slight resistance as the hook enters, then a smooth pull-through."
2. **Sensation.** The crocheter inserts the hook and feels what actually happens. The yarn resists more than expected. The stitch below is tighter than the others. The hook catches on a split strand.
3. **Prediction error.** The difference between prediction and sensation is computed. Something feels off. The resistance is too high — perhaps the hook is entering the wrong part of the stitch, or the tension has drifted.
4. **Update.** The crocheter either adjusts their action (wiggles the hook to find the right insertion point — active inference) or adjusts their expectation (this yarn is just stiffer than usual, the resistance is normal — perceptual inference).

This cycle happens in fractions of a second, below conscious awareness, for every single stitch. The crocheter's agency is not exercised once, at the grand level of choosing a pattern. It is exercised continuously, at the node level, with each insertion of the hook. This is what makes crochet meditative and absorbing: the mind is engaged in a constant, rapid loop of prediction, sensation, and correction.

The meta-level: a neural network designing a neural network. Here is the recursive wonder of it all. The crocheter's brain — a biological neural network of roughly 86 billion neurons — is using its own network architecture to design and build a second network in yarn. The brain's networks encode the pattern, predict stitch outcomes, process sensory feedback from the fingers, and issue motor commands to the hands. The yarn network is the output.

And the yarn network, in turn, feeds back into the brain network. Seeing the fabric take shape updates the crocheter's beliefs about the project. Feeling the texture informs the next prediction. The two networks — biological and textile — are coupled through the Active Inference loop, each shaping the other.

This is not just a curiosity. It is a deep structural truth about agency in Active Inference. Agents are systems that maintain models of their world and act to make the world conform to those models. The crocheter maintains a model of the fabric (the pattern, the expected gauge, the anticipated shape) and acts to make the yarn conform to that model. The fabric, in turn, provides sensory evidence that updates the model. Agent and environment — crocheter and fabric — are coupled through a topological boundary (the Markov blanket of the working edge) in a continuous loop of prediction and action.

4. Collective Neural Architecture

The crocheter-as-architect metaphor extends beyond the individual. In a crochet circle, multiple agents collaboratively build network structure. Consider a group project where each participant crochets a granny square that will be joined into a blanket. Each person is a neural architect designing their own subnetwork (their square), but the final assembly — joining the squares — creates a meta-architecture that no single person designed. The overall network topology of the blanket emerges from the collective decisions of many agents.

In neural network terms, this is **modular architecture**: a network composed of distinct subnetworks (modules) that are separately trained and then connected. Each crochet circle member is training their module (their square) according to shared specifications (the pattern and agreed-upon gauge), and the assembly process links these modules into a larger computational structure.

The Active Inference framing here is particularly rich. Each crocheter has their own generative model. When the squares are joined and some do not quite match — one is slightly larger, another has a different tension — the group must collectively minimize prediction error. Someone blocks the squares to equalize size (active inference on the world). Someone adjusts the joining technique to accommodate variation (active inference on the assembly process). Someone accepts that small differences add character (perceptual inference — updating the model to accommodate the data).

Applications

In crochet, we see agency and neural architecture manifest in:

- **Pattern Design as Architecture Search.** When a crochet designer creates a new pattern, they are performing what machine learning calls "architecture search" — exploring the space of possible network topologies to find one that produces the desired output (the target garment, toy, or artwork). The designer experiments with stitch combinations, counts, and shaping until the fabric matches the vision. This is creative agency expressed through network design.
- **Tension Calibration as Weight Training.** The practice of making gauge swatches and adjusting hook size or tension to match a pattern's specifications is analogous to training — adjusting the weights of a network until output matches the target. Experienced crocheters have well-trained weight parameters; beginners are still in early training epochs.
- **Freeform Crochet as Generative Exploration.** Freeform crochet — where the crocheter works without a pattern, making stitch decisions intuitively — is the fiber arts equivalent of a generative model running in exploratory mode. The crocheter's internal model produces predictions about what might look good, and the crocheter acts on those predictions, observing the result and updating. There is no external target (no pattern), only the agent's own generative model. This is creativity as Active Inference: exploring the landscape of possible fabrics by iteratively predicting and acting.

Conclusion

The crocheter is not a passive executor of instructions. The crocheter is a neural architect — an agent whose brain (a neural network) designs, builds, and continuously refines another network (the fabric) through a stitch-by-stitch process of prediction, sensation, and correction. Stitch selection is activation function choice. Tension is weight assignment. Each hook insertion is a decision node where Active Inference operates in real time. And the recursive beauty of it — a network building a network, an architect made of the same stuff as the architecture — reveals something profound about the nature of agency in Active Inference. We are all, always, networks building networks. The crocheter just happens to do it in a way you can see, touch, and wrap around your shoulders on a cold evening.

In the next module, we turn from how the crocheter builds networks to how the crocheter perceives them: how do your fingers and eyes read the topology of the fabric, and what does that tell us about perception in Active Inference?

Key Terms

Term	Crochet Meaning	Active Inference / Neural Network Meaning
Network architect	The crocheter who designs and builds fabric structure	An agent who selects and constructs computational architectures
Activation function	Stitch type (sc, dc, tr) that determines how input transforms to output	Mathematical function that determines a neuron's output given its input
Weight	Tension — how tightly or loosely the yarn is held and worked	Numerical parameter on a connection that determines signal strength
Decision node	The moment of hook insertion — which stitch, where, how	A point in a process where an agent must select among available actions
Architecture search	Pattern design — exploring stitch combinations to achieve a target fabric	Automated process of finding optimal network topology
Feedforward	Working in rows, one direction	Network where signal flows input-to-output with no loops
Recurrent	Working in the round, end connecting to beginning	Network where output feeds back as input
Fan-out	Shell stitch — multiple stitches worked into one base stitch	One node connecting to many in the next layer (expanding dimensionality)
Fan-in	Decrease — multiple stitches worked together into one	Many nodes connecting to one in the next layer (contracting dimensionality)

Term	Crochet Meaning	Active Inference / Neural Network Meaning
Generative model	The pattern, plus the crocheter's understanding and expectations	Internal model that predicts how sensory observations are generated
Prediction error	Fabric not matching expectations (wrong gauge, unexpected shape)	Discrepancy between predicted and actual sensory input (free energy)
Weight training	Gauge swatch practice — adjusting tension to hit target gauge	Iterative adjustment of network parameters to minimize loss
Modular architecture	Granny squares or motifs made separately and joined	Network composed of distinct subnetworks connected in a larger structure